Solve compound area problems

Use the scaffold, show each step and ask for an adult check when marked.

Name: ____ Date: _________________

1. Read the model: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area. Copy or complete the first step.

2. A rectangle is 14 cm by 9 cm. Find its area and perimeter.

3. Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.

Teacher answers and checking prompts

1. Read the model: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area. Copy or complete the first step.
   Answer 80 cm2.

2. A rectangle is 14 cm by 9 cm. Find its area and perimeter.
   Answer Area 126 cm2; perimeter 46 cm.

3. Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
   Answer 84 cm2.

4. Miro says: Miro uses the sloping side as the height and forgets to halve a triangle's area. Is this correct? Explain.
   Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.

Solve compound area problems

Work independently. Show a complete method and check at least one answer.

Name: ____ Date: __________________

1. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

2. A rectangle is 14 cm by 9 cm. Find its area and perimeter.

3. Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

4. Solve this independently and show every stage: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

5. Use a second method or representation for: A rectangle is 14 cm by 9 cm. Find its area and perimeter.

6. Check this result using an inverse, estimate or known fact: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

7. Prove or explain your reasoning: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.

9. Write a complete sentence explaining how you checked one answer.

10. Write and solve a similar question to: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

Teacher answers and checking prompts

1. A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
   Answer 80 cm2.

2. A rectangle is 14 cm by 9 cm. Find its area and perimeter.
   Answer Area 126 cm2; perimeter 46 cm.

3. Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
   Answer 84 cm2.

4. Solve this independently and show every stage: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
   Answer 80 cm2.

5. Use a second method or representation for: A rectangle is 14 cm by 9 cm. Find its area and perimeter.
   Answer Expected result: Area 126 cm2; perimeter 46 cm. Method or representation may vary.

6. Check this result using an inverse, estimate or known fact: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
   Answer Expected result: 84 cm2. A valid check must be shown.

7. Prove or explain your reasoning: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.
   Answer 60 cm2. The explanation must show why the method works.

8. Identify and correct this misconception: Miro uses the sloping side as the height and forgets to halve a triangle's area.
   Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.

9. Write a complete sentence explaining how you checked one answer.
   Answer Answer should name an inverse, estimate, boundary, known fact or equivalent representation.

10. Write and solve a similar question to: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
    Answer Answers vary. The new question must use the same mathematical structure and include a correct solution.

Solve compound area problems

Prove, compare or generalise. Write enough reasoning for another pupil to follow.

Name: ____ Date: __________________

1. Prove or explain your reasoning: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.

2. Prove the answer to this question, rather than only calculating it: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.

3. Solve this in two different ways and compare them: A rectangle is 14 cm by 9 cm. Find its area and perimeter.

4. Work backwards from the answer to reconstruct the question: 84 cm2.

5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.

6. Create a new example that tests the same idea as: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.

7. Find a second method or representation. Compare its efficiency with your first method.

8. Write one always, sometimes or never statement about today's learning and justify it.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.

Teacher answers and checking prompts

1. Prove or explain your reasoning: Find the area of a triangle with base 15 cm and perpendicular height 8 cm.
   Answer 60 cm2. The explanation must show why the method works.

2. Prove the answer to this question, rather than only calculating it: A shape combines a 10 cm by 6 cm rectangle and a triangle with base 10 cm and height 4 cm. Find total area.
   Answer Expected result: 80 cm2. Proof or justification must match the lesson structure.

3. Solve this in two different ways and compare them: A rectangle is 14 cm by 9 cm. Find its area and perimeter.
   Answer Expected result: Area 126 cm2; perimeter 46 cm. Two valid methods and a comparison are required.

4. Work backwards from the answer to reconstruct the question: 84 cm2.
   Answer One valid reconstruction is: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm. Other equivalent questions are acceptable.

5. Explain why this reasoning fails: Miro uses the sloping side as the height and forgets to halve a triangle's area.
   Answer Mark the perpendicular height, use base times height, and divide by two only for a triangle.

6. Create a new example that tests the same idea as: Find the area of a parallelogram with base 12 cm and perpendicular height 7 cm.
   Answer Answers vary. The example and solution must preserve the mathematical structure.

7. Find a second method or representation. Compare its efficiency with your first method.
   Answer Answers vary. Comparison should refer to accuracy, number structure and clarity.

8. Write one always, sometimes or never statement about today's learning and justify it.
   Answer Answers vary. A valid example and counterexample should support the classification.

9. Create a believable incorrect solution. Identify the exact step where the reasoning breaks.
   Answer Answers vary. The error must be mathematically relevant and the correction must be precise.

10. Change one value or condition in a question. Explain what changes in the solution and what stays the same.
    Answer Answers vary. The explanation must distinguish the mathematical structure from the changed value.